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(54) SIDE TURN INDICATOR INTEGRATED WITH A FENDER OF A VEHICLE

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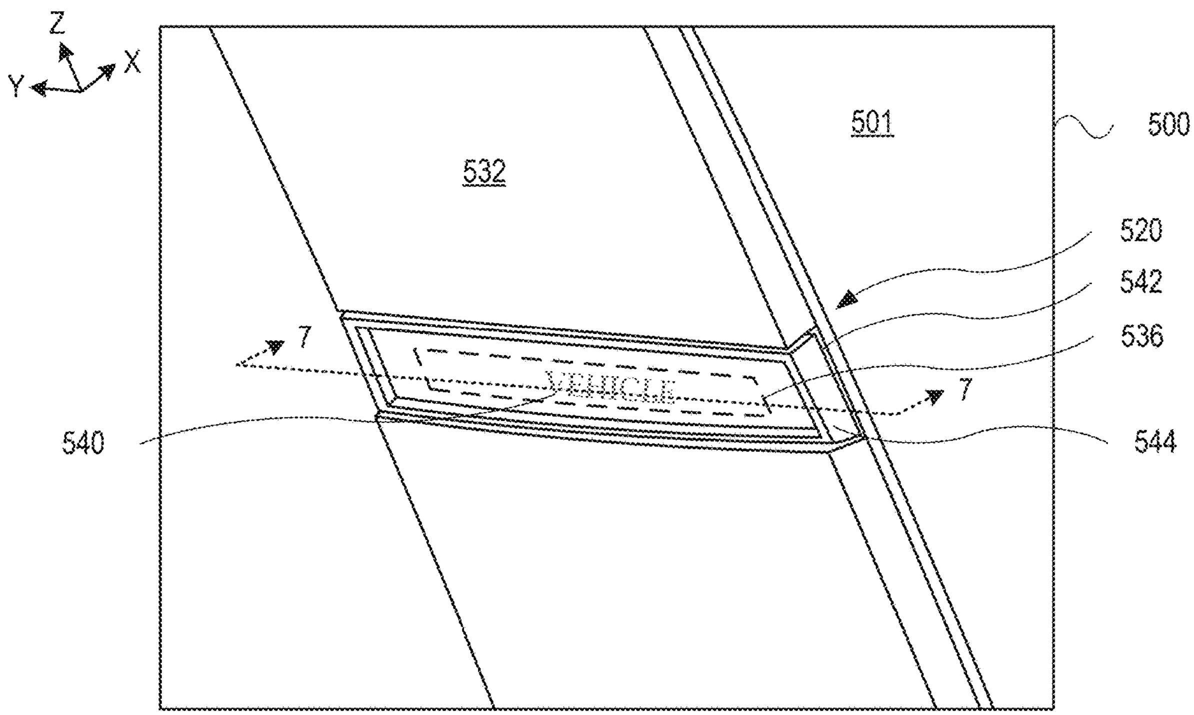
(52) **U.S. Cl.**
CPC ***B60Q 1/38*** (2013.01); ***B60Q 1/326*** (2013.01); ***F21S 43/265*** (2024.05); ***F21S 43/2817*** (2024.05); *F21W 2103/20* (2018.01)

(57) **ABSTRACT**

Vehicle fenders for vehicles include a side turn indicator. A vehicle fender may include a recess portion, and the side turn indicator, including a light source, may be disposed in the recessed portion. The vehicle fender may hide, or otherwise obscure, the light source when the side turn indicator in an inactive mode (e.g., not illuminated). However, when the light source is in an active mode (e.g., the side turn indicator is illuminated), light from the light source is visible through any structure of the side turn indicator and the side turn indicator is visible to present an indication to which the vehicle may turn. Further, the side turn indicator, when illuminated, is generally no longer hidden/obscured by the vehicle fender.



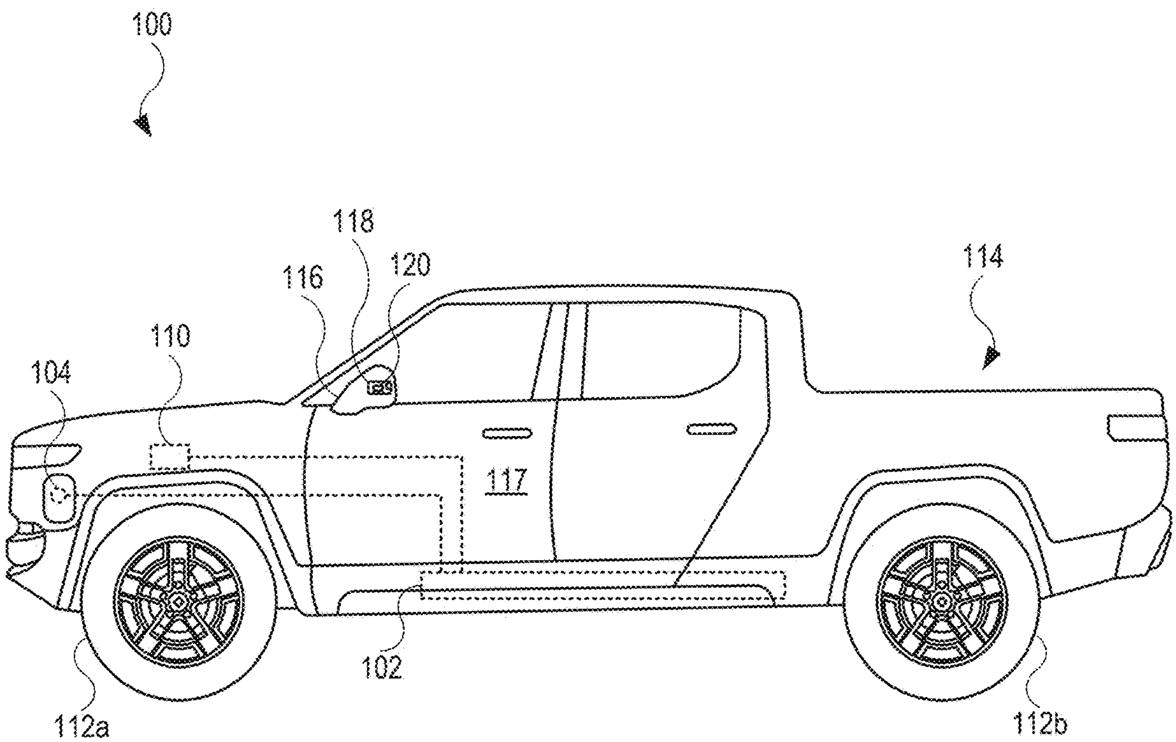


FIG. 1

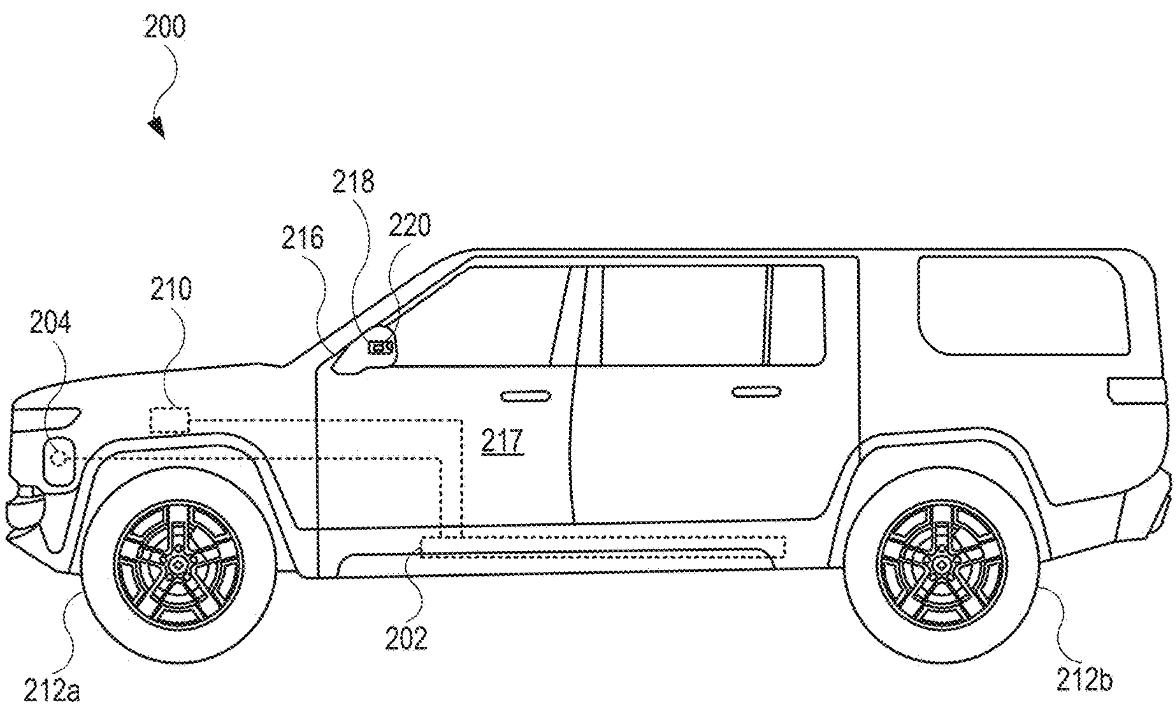


FIG. 2

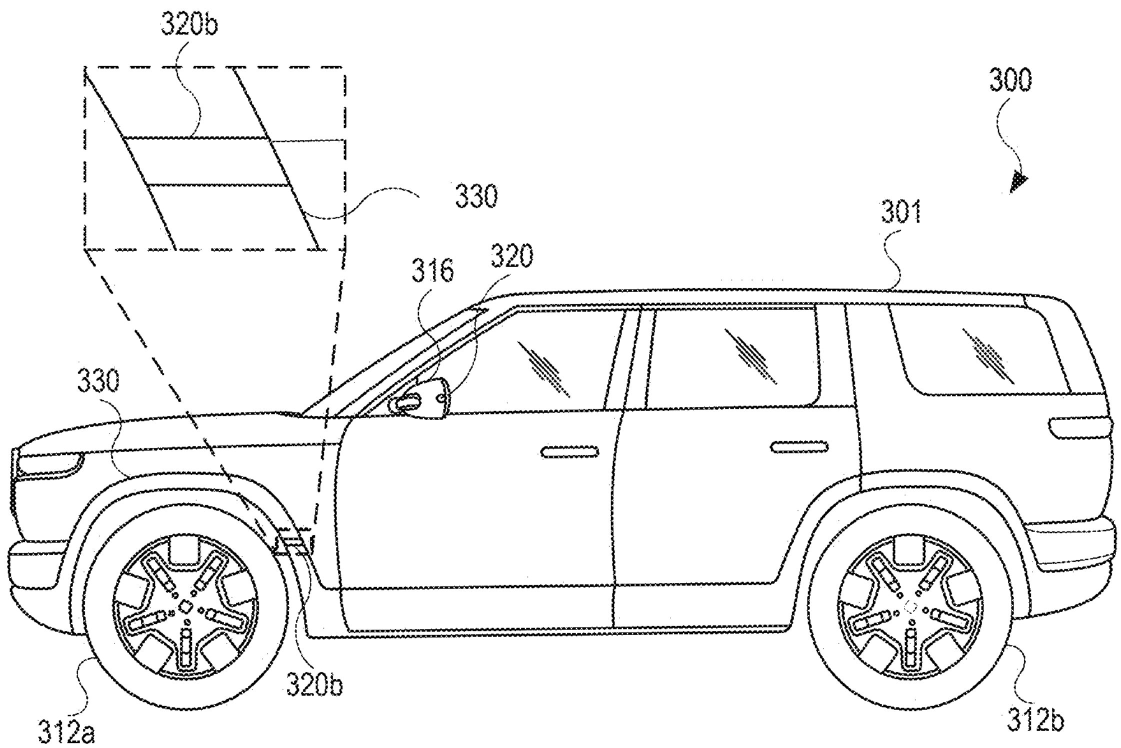


FIG. 3

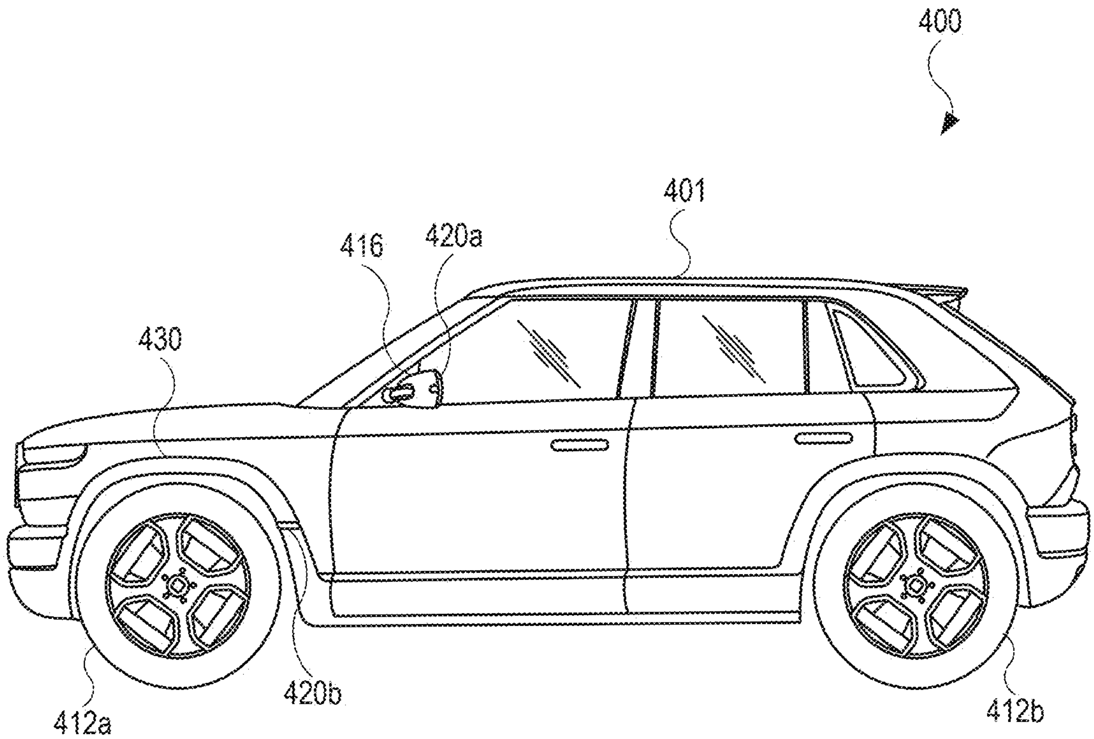


FIG. 4

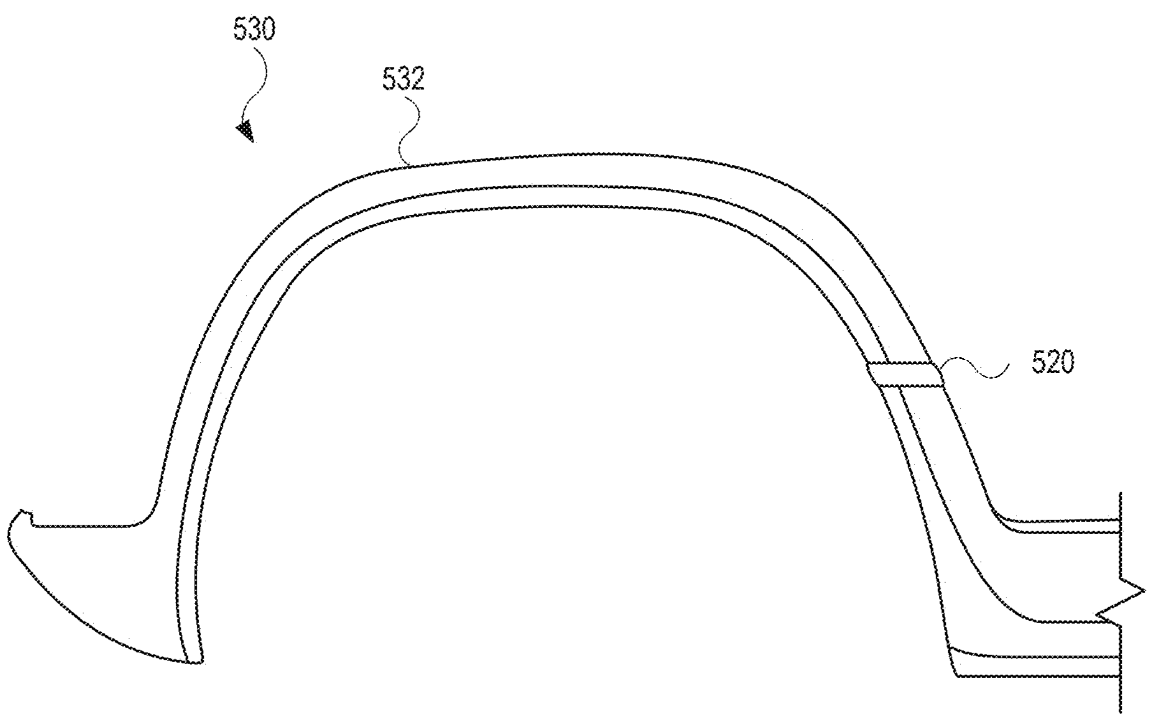


FIG. 5

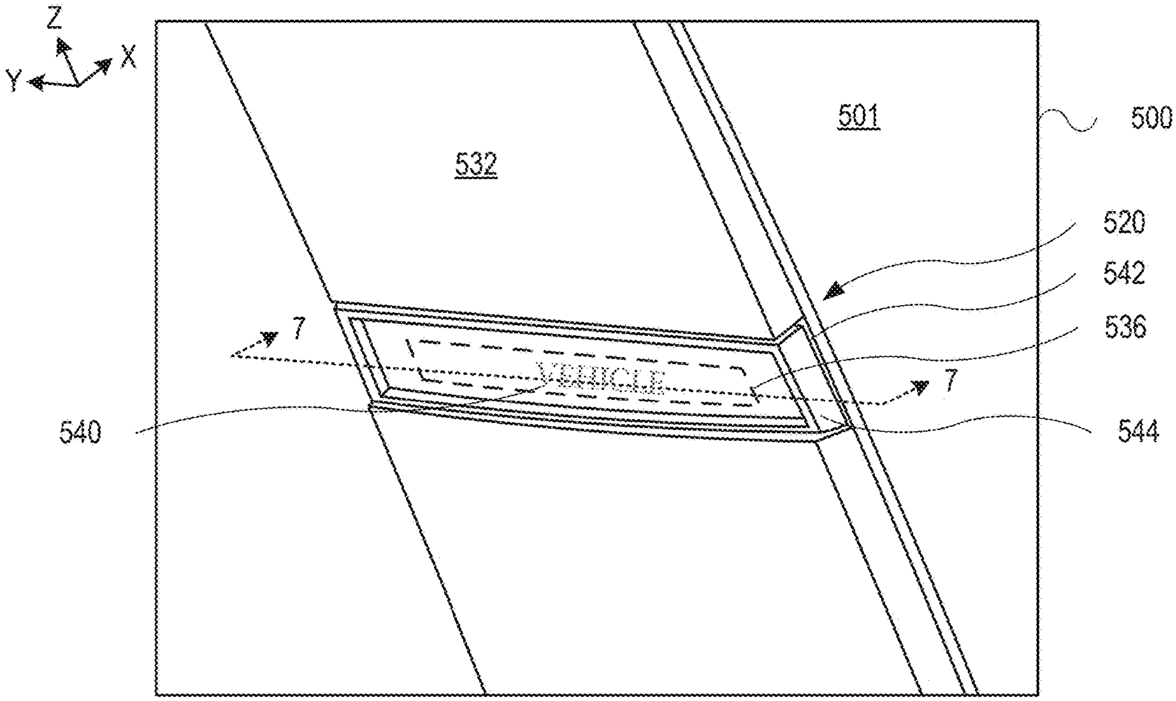

Z
Y
X
501
500
532
520
542
536
7
7
VEHICLE
540
544

FIG. 6A

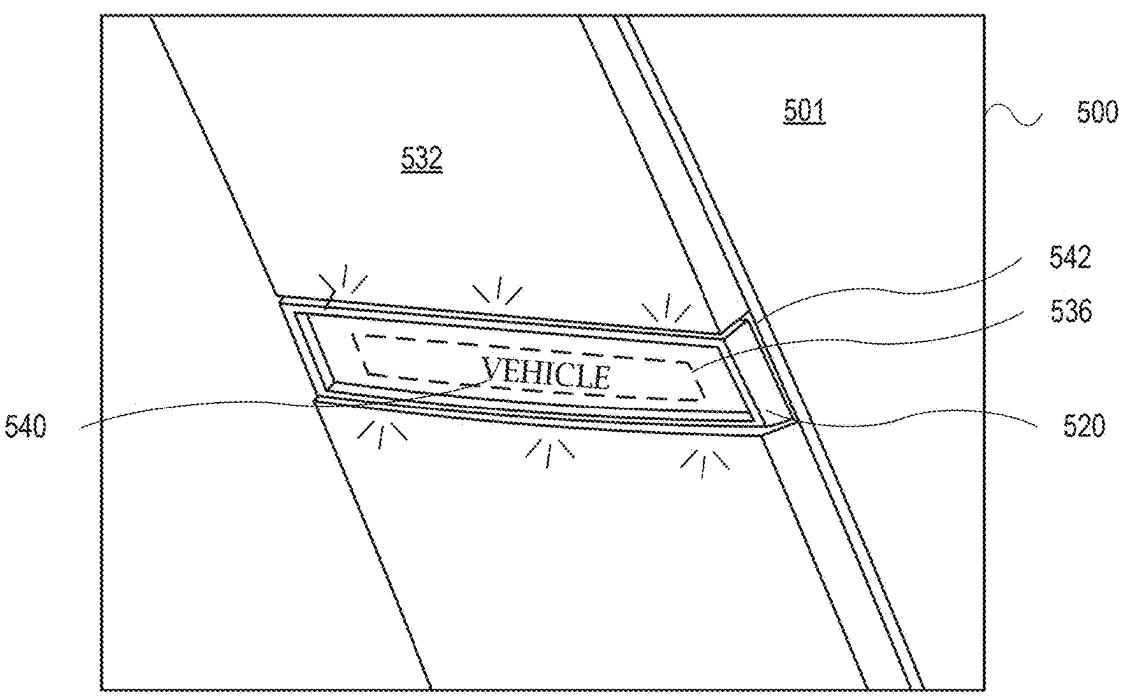

501
500
532
542
536
VEHICLE
540
520

FIG. 6B

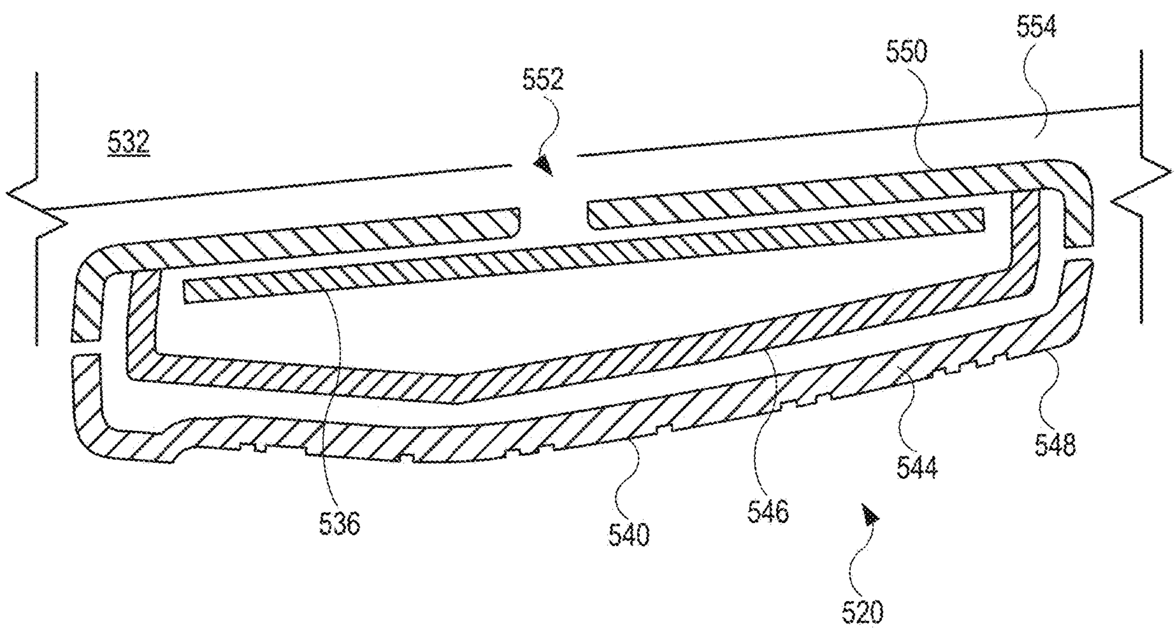


FIG. 7

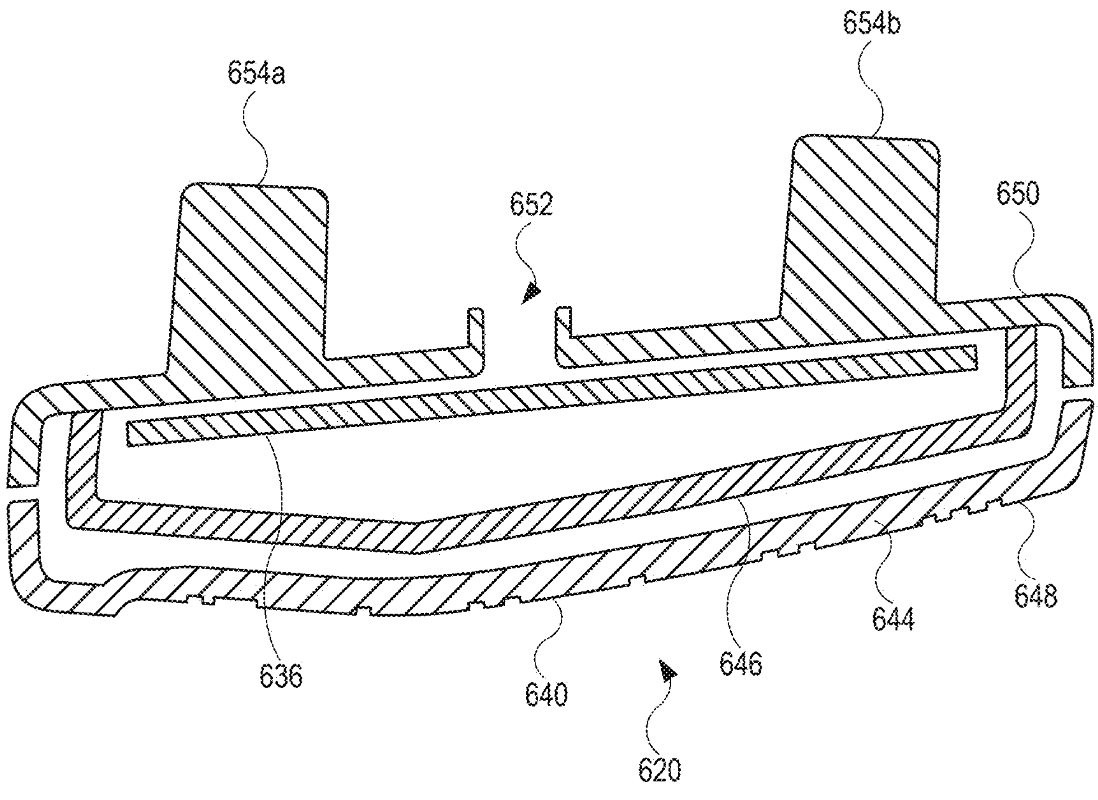


FIG. 8

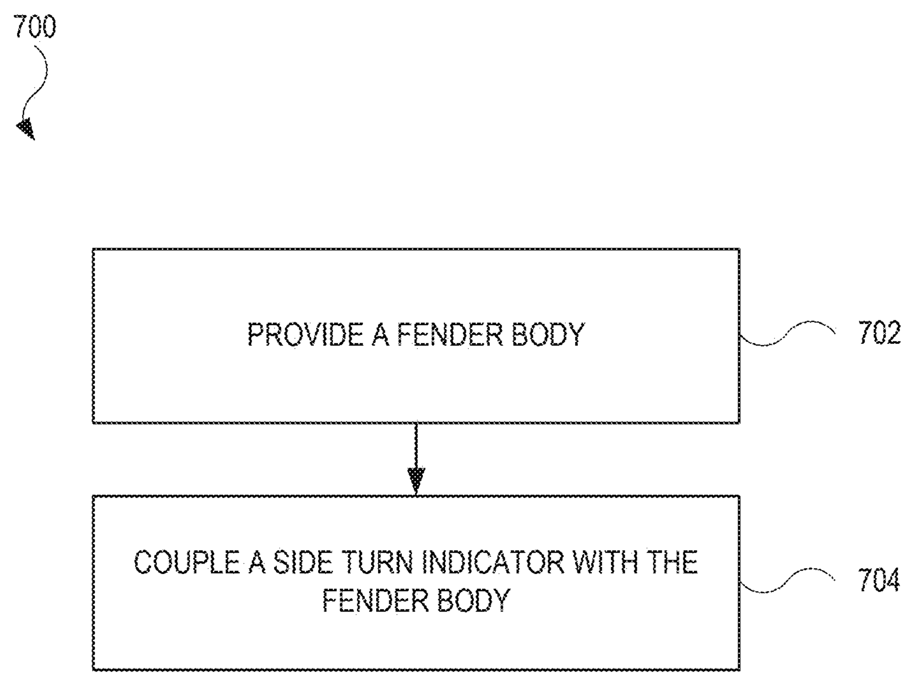

700
PROVIDE A FENDER BODY
702
COUPLE A SIDE TURN INDICATOR WITH THE FENDER BODY
704

FIG. 9

SIDE TURN INDICATOR INTEGRATED WITH A FENDER OF A VEHICLE

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the benefit of priority to U.S. Provisional Application No. 63/562,228, filed on Mar. 6, 2024, titled "TURN SIGNAL INDICATOR INTEGRATED WITH A FENDER OF A VEHICLE", the disclosure of which is incorporated herein by reference in its entirety.

INTRODUCTION

[0002] The present disclosure is directed to vehicles, and more particularly, vehicles with turn signals integrated with fenders positioned on a vehicle.

SUMMARY

[0003] Vehicles may include one or more lamps, some of which are used to provide an indication to drivers in other vehicles as well as to pedestrians. For example, vehicles may include lamps designed to illuminate and provide a side turn indication when the vehicle intends to turn in a direction (e.g., left turn, right turn). In one or more implementations, the one or more light sources are integrated with a vehicle fender of the vehicle.

[0004] In one or more aspects of the present disclosure, a system is described. The system may include a light source on a vehicle body. The system may further include an overlay of the light source. The light source may be configured to direct light to the overlay. The vehicle body may include a vehicle fender. The overlay may include lettering and may be coupled with the vehicle fender. The light source may be configured to direct the light to the lettering. The vehicle fender may further include a recessed portion, and the light source may be disposed in the recessed portion.

[0005] The vehicle fender may include a U-shaped body configured to surround a wheel of a vehicle. The overlay may include lettering. The light source may be configured to direct a portion of the light to the lettering on a vehicle fender.

[0006] The system may further include a lens between the overlay and the light source. The light source may include a side turn indicator for a vehicle. The overlay may include lettering that least partially covers the side turn indicator. The overlay may include a coating. The coating may be configured to at least partially match a color of a vehicle fender.

[0007] The system may further include a housing configured to carry the light source and couple with a vehicle fender. The lens may be positioned between the overlay and the housing.

[0008] In one or more aspects of the present disclosure, a vehicle fender is described. The vehicle fender may include a fender body configured to surround a wheel of a vehicle. The vehicle fender may further include a light source carried by the fender body. The light source may be configured to provide a turn signal for the vehicle.

[0009] The vehicle fender may further include a housing coupled with the fender body. The housing may carry the light source. The vehicle fender may further include an overlay coupled with the housing. The light source may be configured to direct light toward the overlay.

[0010] The vehicle fender may further include a lens. The lens may be positioned between the housing and the overlay. The light source may be positioned between the housing and the lens. The overlay may include an indicium. The light source may be configured to direct light toward the indicium. The indicium may be selected from a group consisting of lettering or a logo.

[0011] The vehicle fender may further include a coating disposed on the overlay. The coating may be configured to at least partially match a color of the fender body. The fender body may include a recessed portion. The light source may be disposed in the recessed portion.

[0012] In one or more aspects of the present disclosure, a vehicle is described. The vehicle may include a wheel. The vehicle may further include a fender that at least partially surrounds the wheel. The fender may include a fender body may include a recessed portion. The fender may further include a light source carried by the fender body at the recessed portion. The light source may be configured to provide a turn signal for the vehicle.

[0013] The vehicle may further include a housing coupled with the fender body. The housing may carry the light source an overlay coupled with the housing. The light source may be configured to direct light toward the overlay. The light source may further include a lens. The lens may be positioned between the housing and the overlay. The light source may be positioned between the housing and the lens.

[0014] Systems, including fenders and/or side mirrors, integrated with a vehicle may include a side mounted light that includes light emission devices, a first lens and a second feature (e.g., lens or transmissive lettering). The side mounted light may include a turn signal or a side marker light. The turn signal may be configured to emit a flashing light when activated by a driver control. The flashing light may be synchronized with at least one of a front and rear turn signal light. The side mounted light may emit an amber color, a 2 Amber color, or a red color. The side mounted light may emit a red light with a minimum photometric intensity of 0.25 (cd)(2) when viewed from a 45 degree left or a 45 degree right test point. The side mounted light may emit an amber light with a minimum photometric intensity of 0.62 (cd)(2) when viewed from a 45 degree left or a 45 degree right test point. The side mounted light may include a mounting height of not less than 15 inches. The side mounted light may include a mounting height of not more than 83 inches from the ground.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] Certain features of the subject technology are set forth in the appended claims. However, for purpose of explanation, several embodiments of the subject technology are set forth in the following figures.

[0016] FIG. **1** illustrates a side view of an example of a vehicle, in accordance with one or more aspects of the present disclosure.

[0017] FIG. **2** illustrates a side view of an alternate example of a vehicle, in accordance with one or more aspects of the present disclosure.

[0018] FIG. **3** illustrates a side view of an alternate example of a vehicle, showing side turn indicators of the vehicle, in accordance with one or more aspects of the present disclosure.

[0019] FIG. **4** illustrates a side view of an alternate example of a vehicle, showing side turn indicators of the vehicle, in accordance with one or more aspects of the present disclosure.

[0020] FIG. **5** illustrates a side view of a fender, in accordance with one or more aspects of the present disclosure.

[0021] FIG. **6**A illustrates an enlarged view of a vehicle, showing a side turn indicator in an inactive mode, in accordance with one or more aspects of the present disclosure.

[0022] FIG. **6**B illustrates an enlarged view of the vehicle shown in FIG. **6**A, showing a side turn indicator in an active mode, in accordance with one or more aspects of the present disclosure.

[0023] FIG. **7** illustrates a cross-sectional view of an example of a side turn indicator shown in FIG. **6**A, taken along line **7**-**7**, showing additional features of the side turn indicator, in accordance with one or more aspects of the present disclosure.

[0024] FIG. **8** illustrates a cross-sectional view of an alternate example of a side turn indicator, in accordance with one or more aspects of the present disclosure.

[0025] FIG. **9** illustrates a flow diagram showing example of a process that may be performed for assembling a system, in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

[0026] The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology can be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the purpose of providing a thorough understanding of the subject technology. However, the subject technology is not limited to the specific details set forth herein and can be practiced using one or more other implementations. In one or more implementations, structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

[0027] The present disclosure is directed to side mounted light sources, such as side turn indicators, integrated with features such as a vehicle fender. Vehicles described herein may include a vehicle fender for each wheel of a vehicle. At least some of the vehicle fenders may include a side turn indicator. In this regard, side mounted lights located on a fender and shown and/or described herein may include a turn signal or a side marker light. The turn signal may be configured to emit a flashing light when activated by a driver-initiated control. The flashing light may be part of one or more flashing lights at the front and rear of the vehicle, with the one or more flashing lights synchronized to flash (e.g., turn on and turn off) at the same time or near the same time. In this regard, the side turn indicator, when in an active mode (e.g., illuminated and flashing), may provide an indication to which direction the vehicle is turning. As non-limiting examples, the side mounted light may emit a color such as amber, 2 Amber, or red. In one or more implementations, the turn signal may be disposed in a recess of the vehicle fender. The side turn indicator may be hidden, or otherwise obscured, by the vehicle fender. The side turn indicator may include a lens that provides an overlay, or cover, to a light source. The lens may be covered or coated with a material, thus shading the lens and changing the appearance of the lens. However, when illuminated, the side turn indicator generates light that passes through lens, including the material, such that the light is visible to drivers and passengers in other vehicles and/or pedestrians near the vehicle.

[0028] FIG. **1** illustrates an example of a vehicle **100**, in accordance with aspects of the present disclosure. In the example shown in FIG. **1**, the vehicle **100** takes the form of a truck. Generally, the vehicle **100** may take the form of any motorized vehicle, including motorized vehicles with an internal combustion engine and/or one or more electric motors. Accordingly, other implementations of the vehicle **100** may include land-based vehicles, such as a car (e.g., sedan, hatchback), a van, or a commercial truck, as non-limiting examples.

[0029] The vehicle **100** may include a battery pack **102**. The battery pack **102** may be coupled (e.g., electrically coupled) to one or more electrical systems of the vehicle **100** to provide power to the one or more electrical systems. The vehicle **100** may further include a port **104** (e.g., charge port) designed to receive a cable connector (not shown in FIG. **1**) used to transmit power (e.g., alternating current (AC) power) that is converted to direct current (DC) power to charge the battery pack **102**. The battery pack **102** may couple to a drive unit **110**, representative of one or more drive units of the vehicle **100**. While the drive unit **110** is shown as generally being in the front of the vehicle **100**, the drive unit **110** may be located in the rear of the vehicle **100**. Further, when multiple drive units are used, at least one drive unit may be in the front of the vehicle **100** to drive the front wheels (e.g., wheel **112***a*), and at least one drive unit may be in the rear of the vehicle **100** to drive the rear wheels (e.g., wheel **112***b*). The drive unit **110** may include, for example, a motor, an inverter, a gear box, and a differential. In the example shown in FIG. **1**, the drive unit **110** takes the form of an electric motor. In this regard, the drive unit **110** may use energy (e.g., electrical energy) stored in the battery pack **102** for propulsion in order to drive (e.g., rotationally drive) wheels of the vehicle **100**. The vehicle **100** may further include a bed **114** that may be used as a storage area for the vehicle **100**.

[0030] Further, the vehicle **100** may include a side mirror **116**. The side mirror **116** may couple with, and extend from, a door **117** (e.g., driver side door representative of a passenger side door) of the vehicle **100**. In one or more implementations, the side mirror **116** takes the form of a driver side mirror. In this regard, the side mirror **116** may provide, based on a reflection from a mirror (not shown in FIG. **1**), an image of objects lateral and/or behind a driver's side of the vehicle **100**. The side mirror **116** may further include an image sensor **118** (e.g., camera) that provide digital images (e.g., still images, motion images) of objects external to the vehicle **100**, such as lateral and/or behind a driver's side of the vehicle **100**. Additionally, the side mirror **116** may include a side turn indicator **120**. The side turn indicator **120** may include one or more light sources (not shown in FIG. **1**) that, when illuminated, provide an indication of travel of the vehicle **100** (e.g., left turn). Although not shown in FIG. **1**, the vehicle **100** may include an additional apparatus that takes the form of a passenger side mirror that include any features shown and/or described for the side mirror **116**, with the additional apparatus provide images of objects lateral and/or behind a passenger's side of the vehicle **100**, and includes one or more light sources that, when illuminated, provide an indication of travel of the vehicle **100** (e.g., right turn).

[0031] FIG. **2** illustrates a side view of an alternate example of a vehicle **200**, in accordance with one or more aspects of the present disclosure. As shown, the vehicle **200** takes the form of a sport utility vehicle (SUV). The vehicle **200** may include several features shown and/or described for the vehicle **100** (shown in FIG. **1**). For example, the vehicle **200** may include a battery pack **202**, a port **204** (e.g., charge port), a drive unit **210** (representative of one or more additional drive units), a wheel **212***a* (representative of an additional front wheel), and a wheel **212***b* (representative of an additional rear wheel).

[0032] Further, the vehicle **200** may include a side mirror **216**. The side mirror **216** may couple with, and extend from, a door **217** (e.g., driver side door representative of a passenger side door) of the vehicle **200**. In one or more implementations, the side mirror **216** takes the form of a driver side mirror. In this regard, the side mirror **216** may provide, based on a reflection from a mirror (not shown in FIG. **2**), an image of objects lateral and/or behind a driver's side of the vehicle **200**. The side mirror **216** may further include an image sensor **218** (e.g., camera) that provide digital images (e.g., still images, video images) of objects lateral and/or behind a driver's side of the vehicle **200**. Additionally, the side mirror **216** may include a side turn indicator **220**. The side turn indicator **220** may include one or more light sources (not shown in FIG. **2**) that, when illuminated, provide an indication the vehicle **200** may turn (e.g., left turn). Although not shown in FIG. **2**, the vehicle **200** may include an additional apparatus that takes the form of a passenger side mirror that include any features shown and/or described for the side mirror **216**, with the additional apparatus provide images of objects lateral and/or behind a passenger's side of the vehicle **200**, and includes one or more light sources that, when illuminated, provide an indication the vehicle **200** may turn (e.g., right turn).

[0033] FIG. **3** illustrates a side view of an alternate example of a vehicle **300**, showing side turn indicators of the vehicle **300**, in accordance with one or more aspects of the present disclosure. The vehicle **300** may include a body **301** (e.g., vehicle body) and a side mirror **316** (e.g., side mirror) coupled with the body **301**. The side mirror **316** may include a side turn indicator **320***a*. Additionally, the vehicle **300** may include a wheel **312***a* and a wheel **312***b*, representative of additional wheels. The vehicle **300** may further include fenders that surround the wheels of the vehicle **300**. For example, the vehicle **300** may include a fender **330** (e.g., vehicle fender) that surrounds, or at least partially surrounds the wheel **312***a*.

[0034] The vehicle **400** may further include a side turn indicator **320***b*. As shown in the enlarged view, the side turn indicator **320** may be integrated with and carried by the fender **330**. In one or more implementations, the vehicle **300** include four wheels and a corresponding number (e.g., four) fenders. Any fender of the vehicle **300** may include a side turn indicator. Also, the side turn indicator **320** may be substituted with other features such as a side marker or a reflector, as non-limiting examples.

[0035] FIG. **4** illustrates a side view of an alternate example of a vehicle **400**, showing side turn indicators of the vehicle **400**, in accordance with one or more aspects of the present disclosure. The vehicle **400** may include a body **401** (e.g., vehicle body). As shown, the body **401** may include a different shape than prior examples of a body of a vehicle. The vehicle **400** may further include a side mirror **416** (e.g., side mirror) coupled with the body **401**. The side mirror **416** may include a side turn indicator **420***a*. Additionally, the vehicle **400** may include a wheel **412***a* and a wheel **412***b*, representative of additional wheels. The vehicle **400** may further include fenders that surround the wheels of the vehicle **400**. For example, the vehicle **400** may include a fender **430** that surrounds, or at least partially surrounds the wheel **412***a*.

[0036] The vehicle **400** may further include a side turn indicator **420***b*. As shown in the enlarged view, the side turn indicator **420***b* may be integrated with and carried by the fender **430**. In one or more implementations, the vehicle **400** include four wheels and a corresponding number (e.g., four) fenders. Any fender of the vehicle **400** may include a side turn indicator. Also, the side turn indicator **420***b* may be substituted with other features such as a side marker or a reflector, as non-limiting examples.

[0037] Referring to FIG. **3** and FIG. **4**, the side turn indicators **320***b* and **420***b* may function as a turn signal for the vehicle **300** and the vehicle **400**, respectively. In this regard, each of the turn signal indicators **320***b* and **420***b* may be configured to emit a flashing light when activated by a driver-initiated control. The flashing light may be part of one or more flashing lights at the front and rear of the vehicle, with the one or more flashing lights synchronized to flash (e.g., turn on and turn off) at the same time or near the same time. As an example, the side turn indicator **320***b* and the side turn indicator **420***b* may flash with the side turn indicator **320***a* and the side turn indicator **420***a*, respectively, at the same time or near the same time. Also, the side turn indicators **320***b* and **420***b*, when in an active mode (e.g., illuminated and flashing), may provide an indication to which direction the vehicle **300** and the vehicle **400**, respectively, is turning. As non-limiting examples, each of the side turn indicators **320***b* and **420***b* may emit a color such as amber, 2 Amber, or red. Alternately, the side turn indicators **320***b* and **420***b* may function as a side marker for the vehicle **300** and the vehicle **400**, respectively.

[0038] In one or implementations, the side mounted light (e.g., a side turn indicator) may emit a red light with a minimum photometric intensity of 0.25 (cd)(2) when viewed from a 45 degree left or a 45 degree right test point. The side mounted light may emit an amber light with a minimum photometric intensity of 0.62 (cd)(2) when viewed from a 45 degree left or a 45 degree right test point. The side mounted light may include a mounting height of not less than 15 inches. For example, each of the side turn indicators **320***a* and **320***b* (shown in FIG. **3**) may be mounted at least 15 inches measured from a reference surface (e.g., surface on which the vehicle **300** is positioned). The side mounted light may include a mounting height of not more than 83 inches from the ground. For example, each of the side turn indicators **320***a* and **320** (shown in FIG. **3**) may be mounted at most 83 inches measured from a reference surface (e.g., surface on which the vehicle **300** is positioned).

[0039] FIG. **5** illustrates a side view of the fender **530**, in accordance with one or more aspects of the present disclosure. The fender **530** may include a fender body **532** that couples with a vehicle body of a vehicle (e.g., body **301** and body **401** shown in FIG. **3** and FIG. **4**, respectively). The fender body **532** may carry the side turn indicator **520**. The fender body **532** may include a U-shaped body. As a result, the fender body **532** may surround, or at least partially surround, a wheel (e.g., wheel **312***a* and a wheel **412***a* shown in FIG. **3** and FIG. **4**, respectively).

[0040] FIG. **6**A illustrates an enlarged view of a vehicle **500**, showing the side turn indicator **520** in an inactive mode, in accordance with one or more aspects of the present disclosure. As shown, the fender **530** is coupled with a body **501** of the vehicle **500**. The vehicle **500** may take the form of any prior example of a vehicle. The side turn indicator **520** may include a light source **536**. In the inactive mode, the light source **536** is off, and accordingly, the side turn indicator **520** is not illuminated. The side turn indicator **520** may further include an indicium **540** (or indicia). In one or more implementations, the indicium **540** takes the form of lettering (e.g., letters), which may form a word or words. As non-limiting examples, the indicium **540** may be applied through operations such as printing or painting. Also, as a non-limiting example, the lettering may spell the name of the manufacturer of the vehicle **500**. The lettering may include transmissive lettering, opaque lettering, or semi-transparent lettering. Alternatively or in combination, the indicium **540** takes the form of a logo representative of the manufacturer of the vehicle **500**.

[0041] Also, the fender body **532** may include a recess **542**. In one or more implementations, the recess **542** includes a dimension (e.g., depth into the fender body along the X-axis of Cartesian coordinates). The dimension may be approximately in the range of 10-20 millimeters (mm). In one or more implementations, the depth is 15 mm. As shown, the side turn indicator **520**, including its components (e.g., the light source **536**), is positioned, or at least partially positioned, in the recess **542** of the fender body **532**. Alternatively, in one or more implementations, the fender body **532** does not include a recess and the side turn indicator **520** is positioned on an external surface of the fender body **532**.

[0042] In the inactive mode, the side turn indicator **520** is hidden, or at least obscured, from view based on one or more factors. For example, the side turn indicator **520** may include an overlay **544**, or lens, that is formed from a transparent, or semi-transparent material, such as plastic. As a result, at least a portion of the fender body **532** may be viewable through the side turn indicator **520** while the side turn indicator **520** is in the inactive mode. Additionally, the side turn indicator **520** may be coated by a material that either renders an opaqueness (e.g., opaque material on the overlay **544**) or a transparent/semi-transparent (e.g., smoke material on the overlay **544**) condition of the side turn indicator **520**. This will be discussed in further detail below. Moreover, based on part on the material, in the inactive mode of the side turn indicator **520**, the indicium **540** is generally hidden or at least obscured.

[0043] FIG. **6**B illustrates an enlarged view of the vehicle **500** shown in FIG. **6**A, showing the side turn indicator **520** in an active mode, in accordance with one or more aspects of the present disclosure. In the active mode of the side turn indicator **520**, the light source **536** is illuminated and the light source **536** is turned on. The active mode of the side turn indicator **520** may include flashing or blinking (e.g., on and off) of the light source **536** at a predetermined frequency. As a result, the light source **536** provides light that passes through the side turn indicator **520**. Moreover, the light from the light source **536** is directed toward the indicium **540**, where the light passes around the indicium **540**. As a result, the side turn indicator **520**, including the indicium **540**, is no longer hidden or obscured. In this regard, the side turn indicator **520**, based in part on the light generated by the light source **536**, and the indicium **540** may be visible. Moreover, the light from the light source **536** is directed toward the indicium **540**, causing contrast between the side turn indicator **520** and the fender body **532**, as well as between the light and the indicium **540**. Accordingly, the indicium **540** may be relatively more visible in the active mode of the side turn indicator **520**. Beneficially, the side turn indicator **520** is hidden, or at least substantially hidden, in the inactive mode, but is readily viewable by others, including other drivers, passengers and/or pedestrians in the vicinity of the vehicle **500**, thus providing an indication the vehicle **500** is turning in a particular direction.

[0044] FIG. **7** illustrates a cross-sectional view of an example of a side turn indicator **520** shown in FIG. **6**A, taken along line **7-7**, showing additional features of the side turn indicator **520**, in accordance with one or more aspects of the present disclosure. The side turn indicator **520** may include a light source **536**, representative of one or more LEDs or incandescent bulbs. The side turn indicator **520** may further include a cover **546**, or lens, that covers the light source **536**. The cover **546** may project and/or direct light from the light source **536** in the active mode. The side turn indicator **520** may further include an overlay **544**. The overlay **544** may include a transparent or semi-transparent body. A material **548**, or coating, may be applied to the overlay **544** to change the appearance of the overlay **544**, and in turn, the appearance of the side turn indicator **520**. The material **548** may include a semi-transparent material (e.g., smoke material) designed to match the appearance of a fender body (e.g., fender body **532** shown in FIG. **6**A) as well as match the appearance of a vehicle body (e.g., body **501** shown in FIG. **6**A). Alternatively, the material **548** may not be present, and the overlay **544** may be formed with a material (e.g., pigment, dye) such that the overlay **544** matches the appearance of the fender body **532** as well as matches the appearance of a vehicle body (e.g., body **501** shown in FIG. **18**). In one or more implementations, indicium **540** is formed (e.g., molded, embossed) into the overlay **544**. Alternatively, or in combination, the overlay **544** may include a material (e.g., opaque material such as ink or paint) that forms the indicium **540**.

[0045] The side turn indicator **520** may further include housing **550** coupled with the overlay **544**. The housing **550** may carry components such as the light source **536** and the cover **546**, and a circuit board (not shown in FIG. **7**). The housing **550** may include an opening **552** to receive, for example, wires and/or flexible circuitry to make electrical connections with the light source **536** and/or the circuit board. The housing **550** may be designed to couple with the fender body **532**. In one or more implementations, when the side turn indicator **520** is coupled with the fender body **532** by one or more of a screw, a fastener, a clip, or the like, a gap **554** may be present between the fender body **532** and the housing **550**. In this regard, air may pass through the gap **554** while the vehicle **500** (shown in FIG. **6**A) is in motion.

[0046] Several relationships among the components of the side turn indicator **520** may be present. For example, the light source **536** may be positioned between the housing **550**

and the cover **546**. Also, the cover **546** may be positioned between the light source **536** and the overlay **544**.

[0047] FIG. **8** illustrates a cross-sectional view of an alternate example of a side turn indicator **620**, in accordance with one or more aspects of the present disclosure. The side turn indicator **620** may include any features shown and/or described for the side turn indicator **520** (shown in FIG. **7**), including a light source **636**, an indicium **640**, an overlay **644**, a cover **646**, a material **648** (applied to the overlay **644**), a housing **650**, and an opening **652** formed in the housing **650**. Additionally, the housing **650** may further include an extension **654***a* and an extension **654***b*, each of which being designed to couple with a fender body of a vehicle (not shown in FIG. **8**).

[0048] FIG. **9** illustrates a flow diagram showing example of a process **700** that may be performed for assembling a system, in accordance with one or more aspects of the present disclosure. For explanatory purposes, the process **700** primarily described herein with reference to the side turn indicators integrated with a fender, including a fender body, shown and/or described in FIGS. **1-8** and the accompanying portions of this detailed description. However, the process **700** are not limited to the apparatuses shown and/or described in FIGS. **1-8**, and one or more blocks (or operations) of the process **700** may be performed by one or more other components of other suitable moveable apparatuses, devices, or systems. Further for explanatory purposes, some of the blocks of the process **700** are described herein as occurring in serial, or linearly. However, multiple blocks of the process **700** may occur in parallel. In addition, the blocks of the process **700** need not be performed in the order shown and/or one or more blocks of the process need not be performed and/or can be replaced by other operations.

[0049] At block **702**, a fender body is provided. The fender body may be coupled, or integrated, with a vehicle body of a vehicle. The fender body may include a recessed portion.

[0050] At block **704**, a side turn indicator is coupled with the fender body. The side turn indicator may be coupled with the fender body at the recessed portion. The side turn indicator may be hidden, or obscured, by, for example, the fender body when the side turn indicator is in an inactive mode. However, the side turn indicator, in an active mode, may be readily visible.

[0051] Systems, including fenders and/or side mirrors, integrated with a vehicle may include a side mounted light that includes light emission devices, a first lens and a second feature (e.g., lens or transmissive lettering. In one or more implementations, the side mounted light takes the form of a side turn indicator. The side mounted light may include a turn signal or a side marker light, as non-limiting examples. The turn signal may be configured to emit a flashing light when activated by a driver control (e.g., through an interface, such as a display or a lever of vehicle). The flashing light may be synchronized with at least one of a front and rear turn signal light. The side mounted light may emit an amber color, a 2 Amber color, or a red color, as non-limiting examples. The side mounted light may emit a red light with a minimum photometric intensity of 0.25 (cd)(2) when viewed from a 45 degree left or a 45 degree right test point. The letters "cd" may refer to candela. The side mounted light may emit an amber light with a minimum photometric intensity of 0.62 (cd)(2) when viewed from a 45 degree left or a 45 degree right test point. In one or more implementations, the side mounted light may include a mounting height of not less than 15 inches. In one or more implementations, the side mounted light may include a mounting height of not more than 83 inches from the ground.

[0052] As used herein, the phrase "at least one of" preceding a series of items, with the term "and" or "or" to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase "at least one of" does not require selection of at least one of each item listed; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases "at least one of A, B, and C" or "at least one of A, B, or C" each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

[0053] When an element is referred to herein as being "connected" or "coupled" to another element, it is to be understood that the elements can be directly connected to the other element, or have intervening elements present between the elements. In contrast, when an element is referred to as being "directly connected" or "directly coupled" to another element, it should be understood that no intervening elements are present in the "direct" connection between the elements. However, the existence of a direct connection does not exclude other connections, in which intervening elements may be present.

[0054] The predicate words "configured to", "operable to", and "programmed to" do not imply any particular tangible or intangible modification of a subject, but, rather, are intended to be used interchangeably. In one or more implementations, a processor configured to monitor and control an operation or a component may also mean the processor being programmed to monitor and control the operation or the processor being operable to monitor and control the operation. Likewise, a processor configured to execute code can be construed as a processor programmed to execute code or operable to execute code.

[0055] Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase (s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

[0056] The word "exemplary" is used herein to mean "serving as an example, instance, or illustration". Any embodiment described herein as "exemplary" or as an "example" is not necessarily to be construed as preferred or advantageous over other embodiments. Furthermore, to the extent that the term "include", "have", or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term "comprise" as "comprise" is interpreted when employed as a transitional word in a claim.

[0057] All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112, sixth paragraph, unless the element is expressly recited using the phrase "means for" or, in the case of a method claim, the element is recited using the phrase "step for".

[0058] The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean "one and only one" unless specifically so stated, but rather "one or more". Unless specifically stated otherwise, the term "some" refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter gender (e.g., her and its) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the subject disclosure.

What is claimed is:

1. A system, comprising:
a light source on a vehicle body; and
an overlay of the light source, wherein the light source is configured to direct light to the overlay.

2. The system of claim **1**, wherein:
the vehicle body comprises a vehicle fender,
the overlay comprises lettering and is coupled with the vehicle fender, and
the light source is configured to direct the light to the lettering.

3. The system of claim **2**, wherein:
the vehicle fender comprises a recessed portion, and
the light source is disposed in the recessed portion.

4. The system of claim **2**, wherein the vehicle fender comprises a U-shaped body configured to surround a wheel of a vehicle.

5. The system of claim **1**, wherein:
the overlay comprises lettering, and
the light source is configured to direct a portion of the light to the lettering on a vehicle fender.

6. The system of claim **1**, further comprising a lens between the overlay and the light source, wherein:
the light source comprises a side turn indicator for a vehicle, and
the overlay includes lettering that least partially covers the side turn indicator.

7. The system of claim **6**, wherein:
the overlay comprises a coating, and
the coating is configured to at least partially match a color of a vehicle fender.

8. The system of claim **6**, further comprising a housing configured to carry the light source and couple with a vehicle fender, wherein the lens is positioned between the overlay and the housing.

9. A vehicle fender, comprising:
a fender body configured to surround a wheel of a vehicle; and
a light source carried by the fender body, the light source configured to provide a turn signal for the vehicle.

10. The vehicle fender of claim **9**, further comprising:
a housing coupled with the fender body, wherein the housing carries the light source; and
an overlay coupled with the housing, wherein the light source is configured to direct light toward the overlay.

11. The vehicle fender of claim **10**, further comprising a lens, wherein the lens is positioned between the housing and the overlay.

12. The vehicle of claim **11**, wherein the light source is positioned between the housing and the lens.

13. The vehicle fender of claim **10**, wherein:
the overlay comprises an indicium, and
the light source is configured to direct light toward the indicium.

14. The vehicle fender of claim **13**, wherein the indicium is selected from a group consisting of lettering or a logo.

15. The vehicle fender of claim **13**, further comprising a coating disposed on the overlay, wherein the coating is configured to at least partially match a color of the fender body.

16. The vehicle fender of claim **9**, wherein:
the fender body comprises a recessed portion, and
the light source is disposed in the recessed portion.

17. A vehicle, comprising:
a wheel; and
a fender that at least partially surrounds the wheel, the fender comprising:
a fender body comprises a recessed portion; and
a light source carried by the fender body at the recessed portion, the light source configured to provide a turn signal for the vehicle.

18. The vehicle of claim **17**, wherein the light source further comprises:
a housing coupled with the fender body, wherein the housing carries the light source; and
an overlay coupled with the housing, wherein the light source is configured to direct light toward the overlay.

19. The vehicle of claim **18**, wherein:
the light source further comprises a lens, and
the lens is positioned between the housing and the overlay.

20. The vehicle of claim **19**, wherein the light source is positioned between the housing and the lens.

* * * * *